Supplemental Material: Reliability-aware active learning for FFLO phase spaces

S1. PHYSICAL MODEL AND FREE-ENERGY FUNCTIONAL

We study a one-dimensional altermagnetic superconductor in the Nambu basis

$$\Psi_k = (c_{k+q/2,\uparrow}, c_{k+q/2,\downarrow}, c^\dagger_{-k+q/2,\uparrow}, c^\dagger_{-k+q/2,\downarrow})^T$$

with

$$H_{\mathrm{BdG}}(k) = \begin{pmatrix} h(k+q/2) - \mu & \Delta_q \\ \Delta_q^* & -\sigma_y h^T(-k+q/2)\sigma_y + \mu \end{pmatrix}. \tag{S1}$$

where $h(k) = t\cos k + (0, \lambda_y \sin k, \lambda_z \sin k + J_A \cos k) \cdot \boldsymbol{\sigma}$. The production code evaluates the normal-referenced occupation-weighted spectral objective

$$F_{\mathrm{impl}}(\Delta, q) = \frac{1}{2N_k} \sum_{k,n} E_n(k;\Delta,q) f\left[\frac{E_n(k;\Delta,q)}{k_B T}\right] + \frac{\Delta^2}{U} - \frac{1}{2N_k} \sum_{k,n} E_n(k;0,q) f\left[\frac{E_n(k;0,q)}{k_B T}\right]. \tag{S2}$$

with $f(z) = 1/(e^z + 1)$ implemented as a stable sigmoid. The sum includes all four BdG eigenvalue branches, the factor $1/2$ accounts for Nambu doubling, and the final term subtracts the $\Delta = 0$ spectrum at the same q. The k sum is divided by the configured full grid size $N_k = 800$. Temperature enters only through the occupations: the production implementation contains no separately added entropy or log-partition term. We therefore use F_{impl} for the implemented objective rather than replacing it by a textbook grand-potential expression. All frozen minima and labels are those obtained from this objective, with energies in units of t.

TABLE S1. Physical and numerical parameters.

Quantity	Value	Role
t	1	energy unit
λ_y, λ_z	0.6, 0.6	spin–orbit couplings
μ	0.55	2D chemical potential
U	1.6	pairing interaction
ω_D	10	momentum mask cutoff
N_k	800	Brillouin-zone grid
coarse Δ grid	$[0, 0.6]$, 400 points	thermodynamic scan
initial q grid	$[-1, 0.5]$, 400 points	momentum scan
$\Delta_\epsilon, q_\epsilon$	$10^{-3}, 10^{-2}$	phase assignment

S2. EXACT BDG ORACLE

Algorithm S1: Thermodynamic and topology evaluation. **Input:** requested coordinate, numerical configuration, and enabled oracle mode.
Output: observables, phase label, numerical status, and applicable topology fields.

1. For each coordinate, construct the BdG spectra on the configured (Δ, q) grid and evaluate the unchanged objective in Sec. S1.

2. Locate candidate minima. In the adaptive-box path, test enclosure; while allowed and necessary, expand the q window directionally (up to the invocation-specific limit). Refine selected global, nearly degenerate, and edge-risk basins using the configured grids (at least 800 q and 600 Δ points in this path).

3. Apply the near-zero-Δ comparison. Assign thermodynamic identity separately from training eligibility and unresolved/rerun flags; incremental scanning reuses previously evaluated momentum intervals in the same robust core.

4. If the topology calculation is enabled and thermodynamic/numerical applicability holds, scan the full-zone gap. Only a resolved gapped state meeting that invocation's applicability conditions receives the Pfaffian index; the online superconducting scope and the FFLO-only main-figure subset are distinguished below; otherwise return the relevant applicability or unresolved status.

5. Return records with numerical metadata. Coordinate reconciliation occurs at dataset merge; it is not a physical classification rule.

Mode and provenance of the conditional branches. The saved Stage-I, II, and III configurations specify `robust_incremental`. In the recovered Stage-III dispatcher this calls the robust adaptive-box core and reuses previously scanned momentum intervals; it is not a different thermodynamic objective. The Stage-IV launch script requests the same mode with basin clustering and selective refinement, at most three refined minima, three optional basins, and a rank-and-cap overflow policy. These are launch-script settings, not proof of every historical shell override. The export does not bind every invocation to an executable hash. The following branch descriptions identify the recovered implementation; unrecorded call-specific settings remain unconfirmed.

For the saved two-dimensional configurations, the condensation-energy ambiguity tolerance is $\epsilon_F = 10^{-6}t$, the strictly-positive-amplitude comparison tolerance is $\epsilon_+ = 10^{-8}t$, and $\Delta_\epsilon = 10^{-3}t$. They serve different purposes. In the robust core, candidate basins are ordered by their refined objective value, not by phase preference; the near-degeneracy window is $\max(10^{-5}t, 5\epsilon_F)$. The dispatcher requests at least 800 momentum and 600 amplitude samples for local refinement. The historical maximum expansion count, absolute momentum bound, and optional basin-selection switches are not fixed by these run JSON files alone and are not inferred from current defaults.

Momentum-window diagnostics and expansion. The recovered core examines the first and last three momentum cells, reduced for very short grids. Edge sensitivity requires a positive-amplitude trace or an objective below $-\epsilon_F$. It flags an edge within $5 \times 10^{-5}t$ of the scanned minimum, a low-energy local minimum near that edge, an outward descent, or an optimum within the recorded edge margin. The left and right flags determine directional expansion; an unbracketed case can request one symmetric expansion. On each affected side the padding is $\max(4|\delta q|, W_q/2)$, clipped at the supplied absolute momentum bound; the new grid preserves or improves the old spacing. No new interval is added when clipping prevents growth. Thus a diagnostic enclosure test, a termination bound, and the returned unresolved flag are distinct quantities.

Near-zero amplitude and returned states. The guard is triggered when enabled and either the best objective is within ϵ_F of zero, the amplitude is within the supplied refinement half-width of Δ_ϵ, or it lies below Δ_ϵ. A separate scan over amplitudes starting at Δ_ϵ provides the stored positive-amplitude comparison. For a normal candidate, a comparison below $-\epsilon_+$ with amplitude above Δ_ϵ replaces the candidate; a value in $[0, \epsilon_+]$ marks a normal-side boundary band. Robust superconducting acceptance requires both amplitude above Δ_ϵ and objective below $-\epsilon_F$; otherwise the recovered branch assigns normal or ambiguous status according to the recorded comparison.

In this core, the returned momentum-unresolved bit specifically requires a superconducting candidate, an optimum at the final edge margin, and exhaustion of the permitted expansions; it is not identical to every edge-risk diagnostic. Nonfinite selected momentum, amplitude, or objective sets a failure bit. A zero status code with no ambiguity is trusted; a normal-side boundary-band record can remain training-eligible without being trusted. Rerun-required is the negation of training eligibility. These implementation distinctions prevent a numerical status from silently becoming a fourth phase. Full per-call binding of optional refinement switches remains a reproducibility limitation.

Cutoff and zero-temperature record. The recovered shared integration kernel uses $|t \cos k - \mu| < \omega_D$ before the momentum sum and retains the configured full-grid denominator. With the stated cutoff and sampled parameter ranges this mask does not remove momentum points. The exported occupation helper evaluates the sigmoid expression directly and contains no explicit zero-temperature branch. A separately version-bound zero-temperature guard was not recovered; no limiting prescription is supplied here as though it were a recorded historical setting. This is an implementation-provenance limitation, not a change to or assessment of the objective above.

Online spectral status versus offline qualification. The recovered online attachment first requires a superconducting thermodynamic candidate, which is broader than the FFLO-only subset displayed in the main figures. It scans $[-\pi, \pi]$ including both endpoints and compares the minimum absolute eigenvalue with $\max(\epsilon_g^{\mathrm{abs}}, \epsilon_g^{\mathrm{rel}} E_*)$. Nonfinite spectral data or an unresolved Pfaffian sign returns unresolved status; a gap at or below the threshold is recorded as gapless, not as a gapped topological phase. Spectral trust additionally requires training eligibility and no momentum/amplitude unresolved bits. The Stage-IV launch script supplies default grid size 2048, relative gap tolerance 10^{-8} and absolute tolerance zero, with environment overrides possible. These online defaults do not certify an offline refined gap. The separately stored topology-pass implementation doubles selected gap grids and locally refines small-gap/small-margin cases; that script targets a different dataset and its settings are not automatically attributed to the Fig.3 endpoint. Its invocation-level applicability remains to be confirmed.

S3. RELIABILITY-STATE HIERARCHY

Accepted/reliable points have an enclosed minimum and a resolved normal-state comparison. Boundary-delicate points lie within the adopted finite energy or order-parameter resolution. A q-window-unresolved point retains low-energy structure at the scanned edge after allowed expansion; a near-zero-Δ-unresolved point lacks a resolved condensation-energy sign. Rerun-required records remain outside the clean physical training view. For the FFLO-only main-figure presentation, topology-inapplicable includes non-FFLO or numerically unreliable states; this display restriction is not the broader online superconducting filter described above, while gap-unresolved states do not receive a $\mathbb{Z}_2$ label. Phase identity, training eligibility, rerun status, and topology applicability are separate fields.

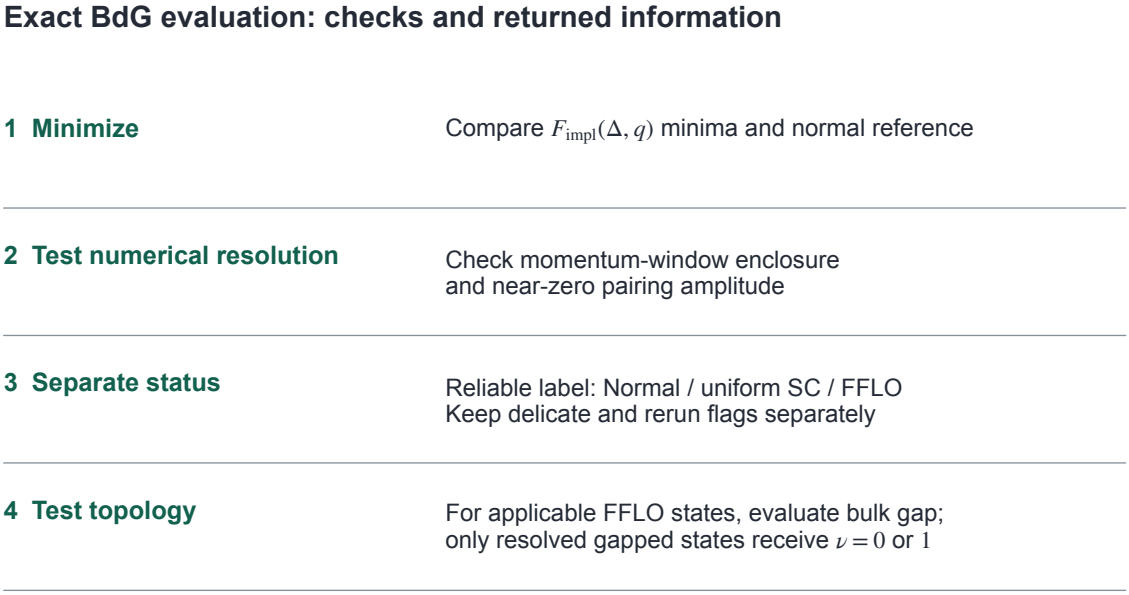


FIG. S1. Exact-oracle reliability-state flow. Numerical status is separate from phase identity, and the main-figure topology subset is restricted to reliable gapped FFLO states.

S4. COMPLETE ACTIVE-LEARNING LOOP

Algorithm S2: Stage-resolved active-learning loop. **Input:** stage configuration and initial or resumed exact dataset (Sec. S6.1).
Output: updated records, trained models, selected-call tables, and diagnostic history.

1. For a cold start, evaluate the stage-specific initial design; for a continuation, load the recorded resume dataset without drawing another seed.

2. At each iteration, construct the eligible training view and phase-stratified training/validation split. Reinitialize and train both ensembles; standardize inputs and regression targets using training statistics.

3. Generate the stage-specific candidate domain, predict observables/probabilities, and score the candidates using Secs. S6.2 and S6.3.

4. Select a batch using Algorithm S3; write the point shards. Evaluate the BdG oracle and preserve failed or unresolved returns as explicit records.

5. Merge the completed shards, reconcile duplicate coordinate keys, and update the eligible learning view without overwriting reliability flags.

6. Recompute the recorded monitoring diagnostics. Continue unless the configured operational stop or budget is reached; save the dataset, configuration, selections, and diagnostics before returning.

S5. SURROGATE ARCHITECTURE AND TRAINING

The implementation uses separate PyTorch ensembles for regression and phase classification, not a Gaussian process or tree model. For Stages I–III, each of five members is an MLP with two 64-unit hidden layers and ReLU activations. Inputs are standardized by the training mean and standard deviation. Regression targets $(\Delta, q, \eta, I_c^+, I_c^-)$ are also standardized; the regression loss is mean-squared error. The classifier uses cross entropy on Normal, uniform-SC, and FFLO labels. Adam uses learning rate 10^{-3} and weight decay 10^{-5}, batch size 512, and 240 epochs for both tasks. Members use seeds $42 + 1000 + i$ and $42 + 2000 + i$ for regression and classification. Models are retrained from scratch at each scheduling step; no early-stopping rule is implemented in these model classes. Regression uncertainty is the member standard deviation; phase probabilities are averaged over members and converted to entropy and margin scores.

TABLE S2. Two-dimensional surrogate and training hyperparameters (Stages I–III).

Item	Value
ensemble size	5 per task
hidden layers	2×64, ReLU
validation fraction	0.15, phase stratified
optimizer	Adam
learning rate / weight decay	$10^{-3}/10^{-5}$
epochs / batch size	240 / 512
regression loss	MSE
classification loss	cross entropy
uncertainty	member dispersion / averaged probabilities

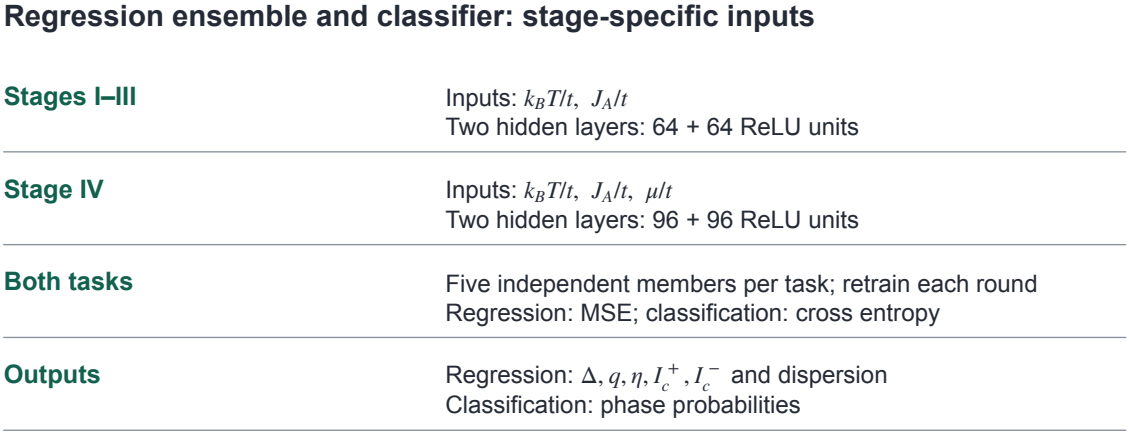


FIG. S2. Stage-specific surrogate architecture. Stages I–III have two inputs and 64-unit hidden layers; Stage IV has three inputs and 96-unit hidden layers. The regression and classification ensembles supply scheduling scores, not physical labels.

S6. ACQUISITION FUNCTION AND MICROBATCH SELECTION

Let $U_{\mathrm{cls}} = 0.4H + 0.6\exp[-m/0.2]$, where H is normalized phase entropy and m is the largest-probability margin. Let $U_{\mathrm{reg}} = (U_\Delta + U_q)/2$, $B_{NS} = \exp[-|\hat{\Delta} - \Delta_\epsilon|/0.04]\, 4P_N P_{SC}$, $B_{UF} = P_{SC}\exp[-||\hat{q}| - q_\epsilon|/0.04]$, and $G = (G_\Delta + G_q)/2$. The Stage-I/II full profile is

$$\begin{aligned} A_{\mathrm{phase}} &= U_{\mathrm{cls}} + 0.6U_{\mathrm{reg}} + B_{NS} + 0.9B_{UF} + 0.5G, \\ A_0 &= A_{\mathrm{phase}} + 0.4E_q + w_{\mathrm{ext}}E_{\mathrm{ext}}. \end{aligned} \tag{S3}$$

where E_q flags predicted q-edge risk and E_{ext} combines extrapolation with uncertainty. w_{ext} decreases piecewise from 0.15 to 0.08 and 0.03 at iterations 10 and 30. High-confidence interiors receive a soft penalty; the active pool uses quantiles of the penalized A_{pool} over eligible hard-unseen candidates, as defined in Sec. S6.2. Observation repulsion uses length 0.02 and floor 0.5. During greedy stochastic construction, batch repulsion uses length 0.03 and floor 0.01, yielding $A_{\mathrm{select}} = A_{\mathrm{pool}}R_{\mathrm{obs}}R_{\mathrm{batch}}$ and sampling probability proportional to $A_{\mathrm{select}}^{\gamma}$, with $\gamma = 1.5, 2.5, 4.0$ across the same schedule. Existing exact coordinates are excluded before selection.

Algorithm S3: Stage-specific repelled selection. **Input:** candidate coordinates, component scores, existing observations, and stage configuration.
Output: ordered selected batch, probabilities, and selection metadata.

1. In 2D discovery, apply exact-coordinate exclusion, interior penalties, and the active-pool threshold; the recent and boundary-band cooldown masks are disabled in this mode. Initialize batch repulsion to one and include observation repulsion.

2. In 3D, use the proposal pool and channel quotas; use the weighted combined score for exhausted-channel backfill. Apply the recorded duplicate handling.

3. While the requested batch is incomplete and positive finite weights exist: form the appropriate score times repulsion, raise to the stage-specific power, normalize, and sample one index.

4. Record its probability, remove the selected index, and multiply the remaining batch factors by $K(d; \ell, r)$. Return early if no eligible weight remains; do not invent extra oracle evaluations.

TABLE S3. Production acquisition components.

Component	Definition	Normalization or weight	Role
phase entropy/margin	U_{cls}	0.4/0.6 mixture	classification ambiguity
regression spread	U_{reg}	min–max normalized; 0.6	phase-facing uncertainty
Normal–SC	B_{NS}	competition gated; weight 1	order-parameter boundary
uniform–FFLO	B_{UF}	SC gated; weight 0.9	finite-q boundary
gradients	G	grid derivative; weight 0.5	rapid variation
numerical risk	E_q	weight 0.4	q-window risk
exploration	E_{ext}	annealed 0.15/0.08/0.03	domain coverage
repulsion	$R_{\mathrm{obs}}R_{\mathrm{batch}}$	length/floor rules	avoid redundant calls

Production acquisition: two-dimensional full profile

1 Score candidates	$A_0 = A_{\mathrm{phase}} + 0.4E_q + w_{\mathrm{ext}}E_{\mathrm{ext}}$ Phase ambiguity, numerical risk, exploration
2 Restrict the pool	$A_{\mathrm{pool}} = A_0 P_{\mathrm{interior}}$ Exact-coordinate masks and scheduled quantiles
3 Select a microbatch	$p(x) \propto [A_{\mathrm{pool}} R_{\mathrm{obs}} R_{\mathrm{batch}}]^{\gamma}$ Update batch repulsion after every selected point
4 Return to BdG	Evaluate selected coordinates; retain numerical status Merge observations and retrain

Stage III uses topology scores; Stage IV uses channel quotas (see text).

FIG. S3. Acquisition decomposition and repelled microbatch selection.

S6.1. Stage-specific settings and auxiliary responses

The run configurations distinguish initialization from continuation. Stage I uses a 512-location random-grid design, seed 42, on the 241×321 temperature–exchange grid. Stage II continues that dataset rather than drawing another independent seed. Stage III starts a separate 512-point scrambled Sobol design, also with seed 42, on the same bounded two-coordinate domain. Its final saved configuration is a resume at iteration 17; the initialization field describes the original run, not 512 additional observations at resume. Stage IV starts 1,024 scrambled Sobol points with seed 20260623 in the three-coordinate domain. The nominal subsequent batch size is 256 in all four stages; actual selected and retained counts need not equal this request.

TABLE S4. Stage-specific training settings extracted from saved run configurations and the corresponding archived model constructors. Optimization batches are distinct from 256-point oracle batches.

Setting	Stages I–II	Stage III	Stage IV
Input coordinates	T, J_A	T, J_A	T, J_A, μ
Hidden layers (ReLU)	$64, 64$	$64, 64$	$96, 96$
Members per task	5	5	5
Epochs (regression/classification)	240/240	240/240	180/180
Optimization batch size	512	512	512
Learning rate	10^{-3}	10^{-3}	10^{-3}
Weight decay	10^{-5}	10^{-5}	10^{-5}
Validation fraction	0.15	0.15	0.15
Base seed	42	42	20260623

All stages train separate regression and classification ensembles. The archived Stage-IV adapter overrides the input dimension, hidden width, epoch count, ensemble size, and seed; its optimization batch size, Adam settings, and validation fraction come from the archived constructor, not from the current development defaults. The target vector remains $(\Delta_{\mathrm{opt}}, q_{\mathrm{opt}}, \eta, I_c^+, I_c^-)$. No additional learned spectral network is used in the Stage-III/IV acquisition implementation examined here:

predicted Δ, q are passed to analytic Pfaffian expressions to construct scheduling proxies. They do not replace exact spectral labels.

Here $I_c^+ \geq 0$ and $I_c^- \leq 0$ are the signed extrema located on the positive and negative current branches outward from the optimized momentum. The implementation records

$$\eta = \frac{|I_c^+| - |I_c^-|}{|I_c^+| + |I_c^-|}, \tag{S4}$$

with zero returned if both currents vanish (and -1 or $+1$ if only the positive or negative branch vanishes). These are auxiliary response targets, not independently validated diode-response claims. In the saved Stage-I/II `full` profile, response scores are exported but excluded from the main acquisition. Stage III's `topo_trivial` profile and Stage IV's explicit three-channel selector likewise use the Δ, q uncertainty channels rather than $\eta, I_c^{\pm}$ to schedule calls. The presence of an optional response-weighted cleanup branch in the archived code is not evidence that it was used in these endpoint configurations.

S6.2. Definitions of the two-dimensional acquisition components

The following definitions apply to the saved two-dimensional implementation. Write $\mathcal{N}[z] = (z - z_{\min})/(z_{\max} - z_{\min})$ over finite values on the candidate array; constant arrays and nonfinite entries map to zero, and the result is clipped to $[0, 1]$. Each uncertainty channel is normalized separately. Grid gradients use first-order edge differences and coordinate spacings, not pixel indices:

$$G = \tfrac{1}{2}\mathcal{N}\left[\sqrt{(\partial_\tau \hat{\Delta})^2 + (\partial_{J_A/t} \hat{\Delta})^2}\right] + \tfrac{1}{2} P_{SC} \mathcal{N}\left[\sqrt{(\partial_\tau |\hat{q}|)^2 + (\partial_{J_A/t} |\hat{q}|)^2}\right]. \tag{S5}$$

Here $\tau = k_B T/t$. Let $[z]_+ = \max(z, 0)$, $d_q = \min(|\hat{q} + 1|, |0.5 - \hat{q}|)$, and $j = J_A/t$. Then

$$E_q = P_{SC} \max\{\exp[-d_q/0.02], \mathcal{N}([j - 1.2]_+)\}, \tag{S6}$$

$$E_{\text{ext}} = \max\{\mathcal{N}([\tau - 0.5]_+), \mathcal{N}([j - 1.2]_+)\} \max\{U_{\text{cls}}, U_{\text{reg}}\}. \tag{S7}$$

The thresholds above are acquisition parameters, not physical phase criteria. The interior indicator requires $P_{\max} > 0.98$ and each of U_{NS}, U_{UF}, G, E_q, and E_{ext} below 0.05, with $U_{NS} = 4P_N P_{SC}$ and $U_{UF} = 4P_{\text{uniform}} P_{\text{FFLO}}$. Its multiplicative penalty is 0.5 before iteration 10 and 0.1 afterward; all other candidates retain factor one. Thus $A_{\text{pool}} = A_0 P_{\text{interior}}$, not the unpenalized score.

The pool quantile is 0.90, 0.95, and 0.98 before iteration 10, from 10 to 29, and from 30 onward. Among eligible hard-unseen candidates, the threshold is the maximum of this quantile and $0.7 Q_{0.95}(A_{\text{pool}})$. The target admitted fraction decreases linearly from 0.20 at iteration zero to 0.05 at iteration 30. If exceeded, the quantile increases in steps of 0.005, up to 0.995, subject to the minimum-batch guard; ties and this quantile ceiling mean this is not an unconditional exact cardinality cap. Before iteration 5, quantile relaxation in steps of 0.02 down to 0.70 may fill the configured minimum batch (64); afterward underfilled batches are allowed and no information-free filling is required.

Distances below use coordinates divided by their domain ranges, $(0.56, 2.12)$, with d_{obs} the nearest retained-coordinate distance. The repulsion kernel and within-batch update are

$$K(d; \ell, r) = r + (1 - r)\{1 - \exp[-(d/\ell)^2]\}, \tag{S8}$$

$$R_{\text{obs}}(x) = K(d_{\text{obs}}(x); 0.02, 0.5), \qquad R_{\text{batch}}(x; \mathcal{B}) = \prod_{s \in \mathcal{B}} K(d(x, s); 0.03, 0.01), \tag{S9}$$

$$p(x \mid \mathcal{B}) = \frac{[A_{\text{pool}}(x) R_{\text{obs}}(x) R_{\text{batch}}(x; \mathcal{B})]^\gamma}{\sum_{z \in \mathcal{P} \setminus \mathcal{B}} [A_{\text{pool}}(z) R_{\text{obs}}(z) R_{\text{batch}}(z; \mathcal{B})]^\gamma}. \tag{S10}$$

Here $\mathcal{P}$ is the eligible active pool and $\gamma = 1.5, 2.5, 4$ on the same iteration intervals. A selected index is removed immediately, and the product is updated after every pick. Existing exact coordinates are excluded using four-decimal coordinate keys; recent-selection and normal-boundary-band cooldowns are disabled by the discovery branch, despite their configuration fields. Repeated requests in historical accounting must not be equated with successful unique returns.

S6.3. Topology-aware and three-dimensional selectors

Stage III replaces the `full` score by $w_p\mathcal{N}(A_{\text{phase}}) + w_s A_s + w_c A_c$, with weights $(0.45, 0.45, 0.10)$ before iteration 30 and $(0.25, 0.65, 0.10)$ thereafter. It does not simply add a new spectral head to the old full score. The spectral proxy is the maximum of a Pfaffian-margin score, an opposite-index proximity score, and a gapless-support proximity score. With normalized distances d_0, d_1, d_g, d_t to trivial, topological, gapless, and trusted-topology observations,

$$A_s = P_{SC} \max\{e^{-m_P/0.02}, e^{-\min(d_0,d_1)/0.06-|d_0-d_1|/0.06}, e^{-d_g/0.06}\}, \tag{S11}$$

$$A_c = P_{SC}\{1 - e^{-(d_t/0.08)^2}\}. \tag{S12}$$

Absent support sets disable their proximity term; absent trusted-topology support gives $A_c = P_{SC}$. The dimensionless margin is $m_P = \min(|P_0|, |P_\pi|)/E_*^2$, with $E_* = \max(1, |t|, |\lambda_y|, |\lambda_z|, |\mu|, |J_A|, |\Delta|)$ in the code's energy units and a 10^{-300} denominator floor. Stage III uses the fixed chemical potential; Stage IV uses the candidate chemical potential. These scores guide selection and confer no topological label.

Stage IV constructs a nominal 65,536-point proposal pool: 70% global scrambled Sobol, 10% jittered local thermodynamic-bracket midpoints, 15% jittered topology-bracket midpoints, and 5% chemical-potential-edge proposals. Missing bracket proposals are backfilled. This is a different proposal strategy from Stage III's diagnostic-only brackets. Coordinates are clipped to the recorded domain and proposal duplicates are removed at eight decimal places. Existing observations are identified at six decimal places. The saved selection tables, not a nominal pool-size assumption, determine actual call counts.

The Stage-IV phase score is $\text{clip}[0.65H + 0.35\,\text{clip}(\bar{\sigma}/Q_{0.95}(\bar{\sigma}), 0, 1), 0, 1]$, where $\bar{\sigma} = (\sigma_\Delta + \sigma_q)/2$ and the denominator has a 10^{-12} floor. Its spectral score takes the larger of the Pfaffian-margin term and the opposite-index term in Eq. (S11), now with distance scale 0.08 and no gapless term. Coverage is $1 - \exp[-(d_{\text{obs}}/0.04)^2]$, without P_{SC}. Distances normalize all three domain ranges $(0.56, 2.12, 2.0)$.

Channel weights $(0.45, 0.40, 0.15)$ change to $(0.25, 0.60, 0.15)$ at iteration 12. The selector allocates integer phase/spectral/coverage quotas from these weights; within each channel it samples the channel score times batch repulsion to power 3. Exhausted channels fall back to the weighted combined score. Batch repulsion uses $\ell = 0.045$, floor 0.02, and the same product update as Eq. (S9). Thus Stage IV must not be described as the two-dimensional quantile-pool selector applied unchanged in three dimensions.

Probability conventions and exclusions. For the three phase probabilities, $P_{SC} = P_{\text{uniform}} + P_{\text{FFLO}}$, $H = -\sum_{a=1}^{3} P_a \log(\max(P_a, 10^{-12}))/\log 3$, and $m = P_{(1)} - P_{(2)}$ with descending probabilities. These quantities are dimensionless. Member predictions are inverse-transformed before their population standard deviation is computed, so σ_Δ and σ_q have the target amplitude and momentum units before the stated score normalizations.

The saved two-dimensional configurations declare a five-iteration recent-selection cooldown and a normal-side boundary-band mask, both using four-decimal coordinate keys. However, the recovered `discovery` branch explicitly clears both masks: they do not exclude candidates in the

runs described here. Exact-coordinate exclusion and soft observation repulsion remain active. In the alternative refinement branch, recent selections come from iterations $\max(0, i-5), \ldots, i-1$; the boundary-band mask is collected from available merged records without this finite lookback. If the requested batch is underfilled, recent and then boundary-band exclusions are relaxed, while exact-coordinate exclusion remains. This distinction also applies to Stage III, whose saved mode is discovery. Its topology score replaces the score composition, while the interior penalty, eligible-pool quantiles, observation repulsion, and sequential batch repulsion remain inherited.

Stage-IV proposal construction and ordered channel selection. The proposal counts are computed by Python rounding of the normalized source fractions times 65,536, with any remainder assigned to global Sobol proposals. The block order is global Sobol, topology-bracket jitter, thermodynamic-bracket jitter, then chemical-potential edges. Opposite-label pairs are sought among eight normalized-coordinate neighbors, retaining only pairs with second index greater than the first and distance at most 0.20. Pairs are sorted by distance and truncated to the requested source count. The topology source uses reliable gapped superconducting support in the recovered adapter, not an implicit FFLO-only filter. Pair indices are sampled with replacement; midpoint jitter is independent Gaussian noise of standard deviation 0.018 times each domain span, followed by clipping.

For edge proposals, the temperature and exchange coordinates come from a Sobol draw. The first half of the chemical potentials are uniform within 0.04 of the domain span above its lower edge, and the remainder within that width below its upper edge. Missing bracket proposals are backfilled with a separate Sobol draw before eight-decimal proposal deduplication; no second fill is promised after deduplication. Proposal randomness uses seed $20260623 + 7717i$, with Sobol offsets $1000 + i$, $2000 + i$, and $3000 + i$ for global, edge and backfill blocks, respectively.

Selection quotas use Python rounding of normalized channel weights times 256; the remainder is added to coverage. All phase-channel picks precede spectral picks, which precede coverage picks. Each pick updates the shared batch-repulsion product. A channel without finite positive weights falls back to the weighted combined score; invalid powered weights trigger the same fallback, and exhaustion terminates the batch. Selection uses seed $20260623 + 1000003i$. These details describe the stored selector and its configuration, not a newly executed reproduction of the selected sequence.

S7. SAMPLING EFFICIENCY

The Stage-I candidate universe is the complete 241×321 Cartesian grid over $k_B T/t \in [0, 0.56]$ and $J_A/t \in [0, 2.12]$, containing 77,361 locations. Its endpoint contains 6,880 unique exact coordinates, so $f_{\mathrm{exact}} = 6880/77361 = 0.08893$ and $R_{\mathrm{call}} = 77361/6880 = 11.24$. Stage II continues on the same candidate domain and contains every Stage-I endpoint coordinate plus 554 new exact coordinates, producing the 7,434-coordinate reliability endpoint displayed in main-text Fig. 2. Shared coordinates have identical phase labels. Selection records exceed endpoint coordinates because repeated requests and selected coordinates without a retained endpoint return are separate accounting layers. A boundary-enrichment factor cannot be reconstructed because complete candidate-pool boundary labels are absent. Per-iteration timings exist, but no matched exhaustive-grid baseline supports a wall-time speedup.

TABLE S5. Stage-resolved selection and retained-coordinate accounting.

Quantity	Stage I	Stage II	Meaning
prescribed candidate locations	77,361	77,361	common Cartesian domain
selected call records	8,192	9,216	requests, including repeats
unique selected coordinates	7,305	7,885	deduplicated requested locations
unique exact endpoint coordinates	6,880	7,434	retained exact observations
training-eligible records	6,880	7,434	clean learning view
trusted exact records	6,104	6,598	reliable subset
selected without endpoint record	425	451	no retained endpoint return
sampling fraction	8.893%	–	Stage-I discovery headline
exact-location reduction	11.244×	–	grid locations, not wall time

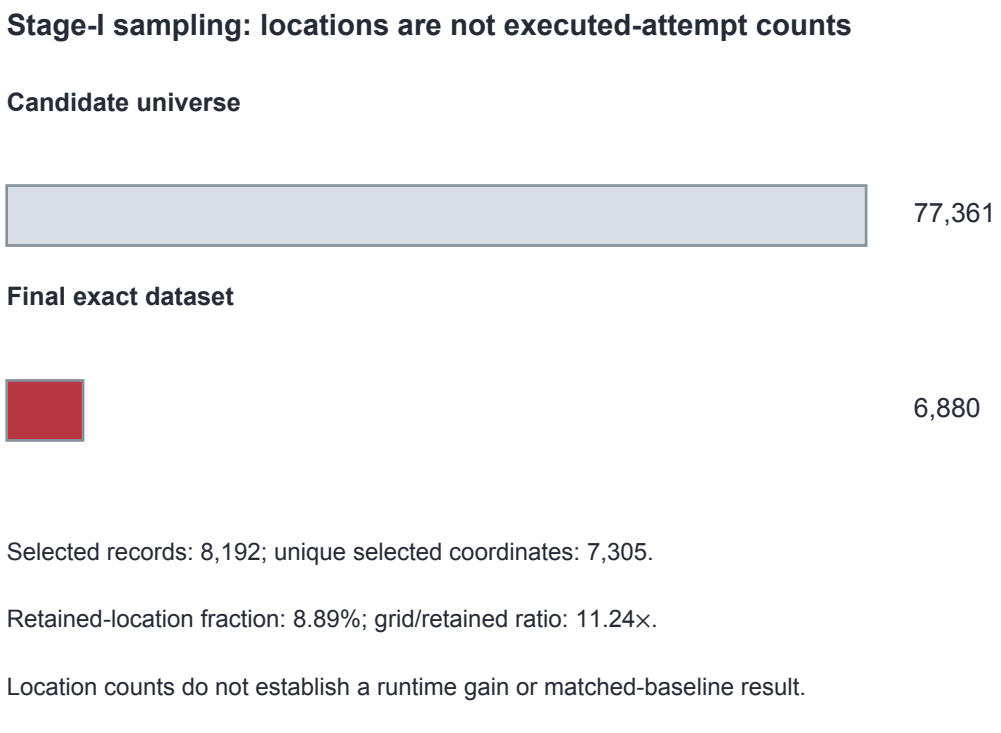


FIG. S4. Stage-I candidate-universe and exact-location accounting. The reduction is a location-count ratio, not a wall-time acceleration.

S8. NUMERICAL DIAGNOSTICS AND OPERATIONAL STOPPING

These criteria are operational controls for the numerical workflow. They are not thermodynamic observables or universal physical scales. Phase-map change is the fraction of fixed monitor-grid cells whose predicted phase changes between iterations. Boundary displacement is a bidirectional nearest-neighbor p_{95} distance between successive extracted boundary point sets after normalization by the parameter-domain ranges. Coverage is the p_{95} distance from the current monitor-grid boundaries to exact samples. Surprise is the fraction of returned labels that differ from the prior prediction, with trusted and flagged denominators stored separately. Four of five main conditions

must hold for four successive iterations after a five-iteration minimum. The adopted thresholds reflect the monitor-grid resolution and workflow tolerance, not a physical scale.

TABLE S6. Two-dimensional online thermodynamic operational diagnostics; not the Stage-III offline topology or Stage-IV controller.

Diagnostic	Definition	Domain	Operational setting	Interpretation
phase-map change	changed-cell fraction	fixed 241×321 monitor	< 0.002	map stability
Normal–SC shift	bidirectional p_{95}	normalized domain	< 0.004167	boundary stability
SC–FFLO shift	bidirectional p_{95}	normalized domain	< 0.004167	boundary stability
coverage	boundary-to-exact p_{95}	normalized domain	< 0.00625	sampling support
label surprise	prediction/return mismatch	selected calls	< 0.05	sequential stability
patience	successive eligible iterations	iteration history	4	operational termination

Stage-resolved stopping records. The thermodynamic five-condition controller above applies to the two-dimensional online loop; it is not the three-dimensional topology controller. Stage I ends with saved reason `max_iterations` at online iteration 30, whereas Stage II ends with `converged_main_phase_boundaries` at 34. Stage III records that same thermodynamic stop reason at online iteration 17; the separate offline topology audit assesses the cumulative endpoint numbered 18. It uses a 401×401 grid, map-change tolerance 0.002, normalized boundary-shift tolerance $1/240$, coverage tolerance 0.00625, and trusted-topology surprise tolerance 0.02. The reported $k_{\mathrm{nn}} = 8$ result satisfies that audit; $k_{\mathrm{nn}} = 6$ marginally exceeds its coverage setting. These are separate from the online four-of-five patience rule. Stage IV has a configured budget of 24 acquisition batches; its saved three-dimensional diagnostics are not evaluated with Table S6's thermodynamic thresholds. A complete job-exit record binding budget exhaustion to every saved endpoint is not established here.

Stored two-dimensional reconstruction algorithm. The topology audit uses inverse-square-distance averages of the saved P_0 and P_π values over k_{nn} neighbors, with distances normalized by the parameter spans and a 10^{-9} distance floor. Predictions are supported only within distance 0.045 of a training point. The final audit mask additionally intersects this support with the convex hull of the final training coordinates. Here k_{nn} counts neighbors and is not crystal momentum k.

Zero contours of each reconstructed Pfaffian field are extracted only in the supported mask. Segments shorter than normalized arc length 0.0015 are excluded. A retained segment needs at least 75% valid vertices and at least 10% vertices supported either by both index classes within 0.045 or by a local bracket midpoint within 0.03. Brackets join distinct opposite-index neighbors separated by at most 0.03; duplicate unordered pairs are removed. They supply support diagnostics, not exact boundary locations or additional oracle calls. This describes the already stored reconstruction; no contour was re-extracted for this revision.

Figure 2(d)'s dashed references are the saved thermodynamic controller settings. Figure 3(c)'s dashed references belong to the separate topology audit just described; panel (d) compares the neighbor-count sensitivity against its coverage tolerance. Figure S5 has no threshold lines. Its last three displayed map-change, displacement, and surprise values increase rather than demonstrate

monotonic stabilization. The three-dimensional monitor reconstruction and all invocation-specific termination details still require a fully bound source-and-run record; the two-dimensional reconstruction algorithm must not be silently substituted for them.

S9. TOPOLOGY DIAGNOSIS AND RECONSTRUCTION SENSITIVITY

Topology is applicable only to thermodynamically accepted, numerically reliable, gapped FFLO states. The minimum absolute BdG eigenvalue is scanned over the full Brillouin zone. At $k = 0, \pi$, the analytic Pfaffians are

$$P_0 = (\mu - t\cos(q/2))^2 + \Delta^2 - (\lambda_y \sin(q/2))^2 - (J_A \cos(q/2) + \lambda_z \sin(q/2))^2, \tag{S13}$$

$$P_\pi = (\mu + t\cos(q/2))^2 + \Delta^2 - (\lambda_y \sin(q/2))^2 - (J_A \cos(q/2) + \lambda_z \sin(q/2))^2. \tag{S14}$$

Away from the relative Pfaffian tolerance 10^{-8}, $P_0 P_\pi < 0$ gives $\nu = 1$ and $P_0 P_\pi > 0$ gives $\nu = 0$. Analytic values were checked against direct Majorana-basis Pfaffians, and CPU/GPU full-zone gaps were compared in the archived topology validation. The displayed transition contour is a deterministic support-restricted reconstruction on a 401×401 monitor grid with support radius 0.045. For neighbor counts $k_{\mathrm{nn}} = 6, 8, 12$, its arc lengths are 0.8270, 0.8215, and 0.8038, while coverage p_{95} is 0.006269, 0.006185, and 0.005913. This sensitivity and the local opposite-$\mathbb{Z}_2$ brackets make the contour a guide to the gap-closing region rather than an exact boundary; nodal alternatives are not exhaustively excluded.

S10. THREE-DIMENSIONAL EXTENSION

Stage IV samples $(k_B T/t, J_A/t, \mu/t)$ with $\mu/t \in [-0.5, 1.5]$. The main figure shows the exact thermodynamic point cloud, reliable FFLO topology support, and bins centered at $\mu/t = -0.45, 0.15, 0.85$, and 1.45 with half-width 0.025. No interpolation is performed between these bins. Archived map-change, proxy-surface displacement, coverage, surprise, and component-count histories are retained as numerical diagnostics only. The discrete support reveals systematic three-dimensional structure but does not establish a converged continuous topology surface or complete atlas.

TABLE S7. Main-text Fig. 4(c) bin accounting. Each window has half-width 0.025. Trusted here is the numerical-reliability subset, not the gapped-FFLO topology subset.

μ/t center	Retained	Normal	Uniform SC	FFLO	Trusted
−0.45	272	95	1	176	255
0.15	154	53	4	97	149
0.85	223	57	18	148	208
1.45	142	104	3	35	139

Main-text Fig. 3(a) contains 3,213 trusted gapped FFLO observations (2,290 cFFLO and 923 tFFLO). The main-text reconstruction retains the $k_{\mathrm{nn}} = 8$ operational-audit pass and the $k_{\mathrm{nn}} = 6$ sensitivity deviation; neither is a universal physical convergence criterion. The four three-dimensional bin totals in Table S7 count retained observations, not estimates of phase-space volume.

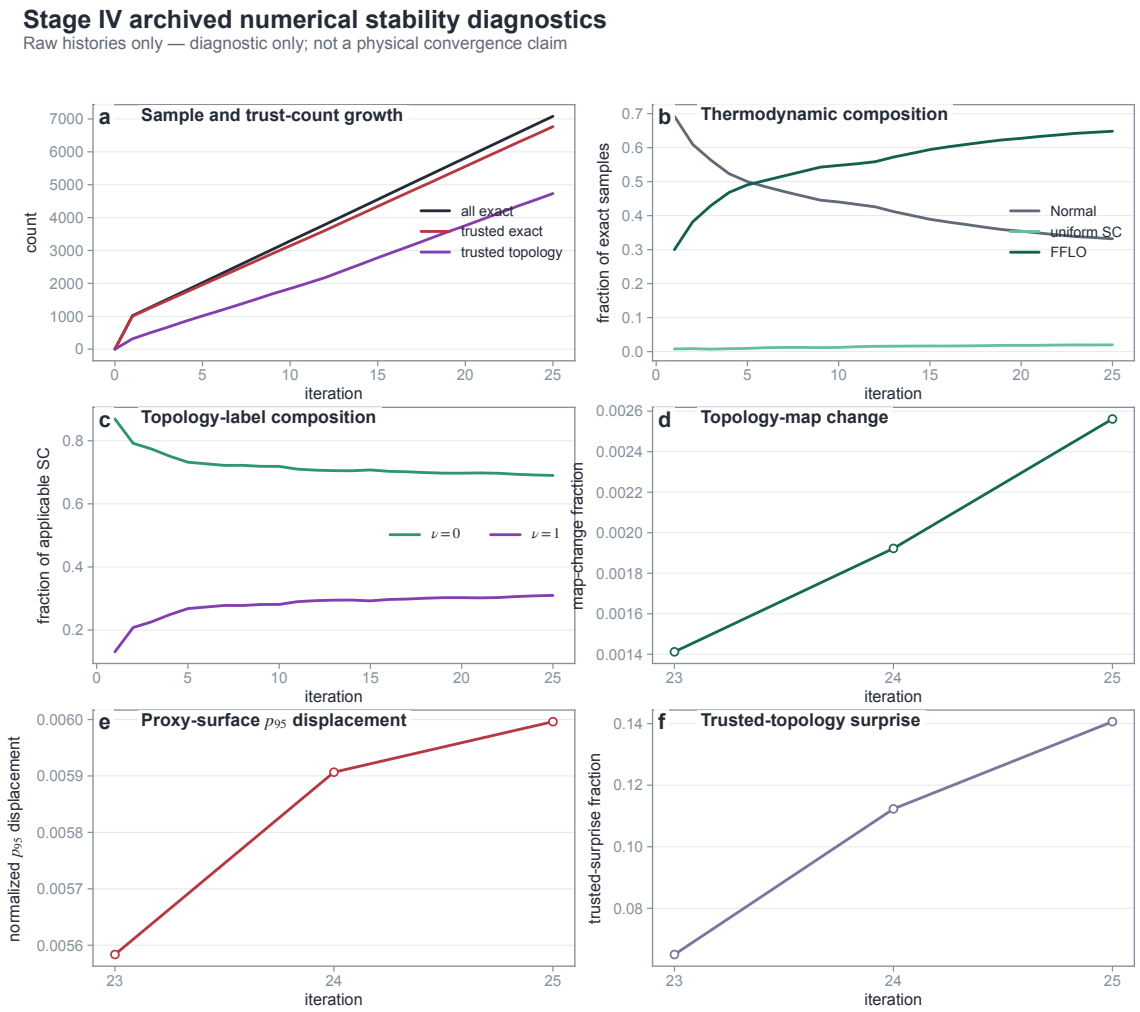


FIG. S5. Stage-IV numerical histories, without threshold lines. (a) Retained and trusted sample growth. (b) Thermodynamic composition. (c) Topology composition. (d) Available topology-map changes. (e) Proxy-surface displacement. (f) Trusted-label surprise. These are iteration diagnostics, not a formally converged three-dimensional topology surface.

S11. REPRODUCIBILITY, CODE, DATA, AND ENVIRONMENT

The scientific entry points and the manuscript build serve different purposes. The historical two-dimensional run configuration identifies `scripts/slurm_exact_oracle_array.sh` for pointwise jobs; active-learning preparation is implemented in `ml_phase/active_refine.py`. The corresponding archived modules include `ml_phase/exact_oracle.py`, `models.py`, `acquisition.py`, and `topology_oracle.py`. The three-dimensional launch package contains `scripts/submit_stageiv_3d_full_loop.sh`, `scripts/stageiv_3d_select.py`, and `scripts/slurm_stageiv_exact_array.sh`; selection calls `ml_phase/stageiv_3d.py` with the saved production configuration. These identify scientific entry points, not a claim that running a LaTeX script repeats the physical calculations.

The export contains archived source snapshots and run configurations, not a public executable service. Stages I–III record Python 3.10.20 and PyTorch 2.11.0+cu130, with eight round-robin oracle shards. These environment and deployment entries should not be generalized to all runs: the

Stage-IV launch script defaults to eight cost-aware shards and CPU selection, but shell overrides and the complete deployment environment are not established by the production JSON alone. Reproduction additionally requires the launch package, actual job overrides, and platform setup. Bitwise GPU reproducibility is not established by a fixed pseudorandom seed alone.

The retained datasets, selection tables, and numerical histories underlie the main figures. In particular, main-text Fig. 2 uses the Stage-II retained-coordinate dataset, Fig. 3 uses the reliable gapped FFLO subset and support-restricted reconstruction, and Fig. 4 uses the Stage-IV point cloud with the bin convention in Sec. S10. Table S5 separates requests, unique requested locations, and retained observations. Total executed evaluations, including failed attempts and repeat work, cannot be inferred from these three counters alone; no complete executed-attempt ledger is claimed here.

The accompanying document-source package contains a separate `build.py` that regenerates the double-column main text, single-column review text, and this Supplement from a clean temporary directory. This script only builds documents from unchanged figure assets. Input hashes identify the source versions because the local export contains no Git repository metadata. Neither the archived scientific package nor these manuscript sources have a confirmed public repository URL or persistent identifier; public access and long-term deposition remain to be arranged by the authors.

Saved run configurations establish the stage-specific parameter choices, while the export provides shared and stage-specific source snapshots. A per-run commit or executable hash linking every Stage-I/II invocation to the shared source snapshot is not available in this package. Accordingly, the method descriptions document the recovered implementation and recorded configurations; they are not a demonstrated end-to-end rerun or a complete executed-attempt ledger.